

Cold pressed corn (*Zea mays*) oil

Zarina Mushtaq^a, Muhammad Imran^b, Nazir Ahmad^b, Muhammad Kamran Khan^b, and Nosheen Asghar^a

^aDepartment of Food Science, Nutrition and Home Economics, Government College University, Faisalabad, Pakistan, ^bInstitute of Home and Food Sciences, Faculty of Life Sciences, Government College University, Faisalabad, Pakistan

1 Introduction

The cold press method for oilseeds and kernels is currently gaining attention for its special oil. There are supercritical extraction and solvent extraction methods in addition to regular expeller pressing to obtain oil, but each technique has its own advantages and drawbacks. The most important differences among the extraction techniques are related to oil yield and the minor component compositions of the oil. Thereby, oils extracted by different techniques usually have different end points for use; some must be fully refined, while some could be used directly for edible, cosmetic, or other purposes.

Corn is primarily grown for its starch and protein, while corn oil (extracted from corn germs) is a by-product of processing. Corn is composed of 61%–78% starch, 6%–12% protein, and 3.1%–5.7% oil. During the dry or wet milling process, corn germ as oil source is separated from the kernel for starch production. A wet milling process is usually preferred for corn germ separation to extract the oil. Corn oil is usually extracted with expeller pressing together with hexane or isohexane extraction to achieve around 97%–99% yield. Refining of crude corn oil is usually carried out by alkaline refining, bleaching, dewaxing, and deodorization processes (Stan, 2015). Corn oil is highly rich in essential fatty acids like omega-6 and omega-9. The omega-6 content was found to be 52.6%. Corn oil effluent is a source of polyunsaturated lipids (53.8%) and saturated lipids (14.8%) (Carrillo et al., 2017).

A comparison was made between cold pressed and fully refined winterized corn oils. The results showed that cold pressed corn oils have more pronounced free fatty acidity (FFA), peroxide (PV), and *p*-anisidin (*p*-AV) values, and higher saponification number, total carotenoids, and phenolic contents than those of the refined winterized corn oils. Some phytoosterols (β -sitosterol, campesterol, and stigmasterol) were analyzed in cold pressed corn oil as well as various flavonoids (hesperidin and rutin) and phenolic acids (gallic, syringic, rosmarinic, and *trans*-ferulic) were detected in the cold pressed oil samples (Güneşer, Yilmaza, & Ok, 2017). Monounsaturated fatty acids (MUFAs):polyunsaturated fatty acids (PUFAs), saturated:unsaturated, as well as linoleic:linolenic acid ratios were improved on receiving a 75% recommended dose of fertilizer (Ray, Banerjee, Dutta, Hazra, & Majumdar, 2019).

Cold pressing is an easy, rapid, environmentally friendly, and cheap process, which yields good-quality oil but it is poor in overall oil yield. Hence, it is only preferable when special purpose oils are demanded. Furthermore, there are strict requirements for producing an acceptable grade of cold press oil. First, the seed or kernel must be very clean, homogeneous, and free from all unacceptable contaminations. The extraction process must be under mild conditions (no heating, solvent, or other chemicals). Finally, the cold pressed oils must be of edible grade regarding physicochemical and sensory properties, and must be safe. Under these circumstances, cold pressed oils could be very beneficial for human health, and could have exceptional aroma. Cold pressed oils do not contain traces of hexane or other solvents, and retain most of their bioactive components (tocopherols, sterols, phenols, etc.), making them highly valued products (Khoddami, Che Man, & Roberts, 2014; Yilmaz, Aydeniz, Güneşer, & Arsunar, 2015).

2 History of corn oil

Corn or maize (*Zea mays* L.) botanically belongs to the grass family gramineae (Poaceae). The cultivation of this plant started in Mesoamerica, and it is currently grown worldwide. The major corn producing countries are America, China, Brazil, Mexico, Indonesia, India, France, and Argentina. In 2016, 1.01 billion metric tons of corn were produced and more than 159 million hectares were used for corn cultivation. Production is increasing each year (Anonymous, 2015).

3 Fatty acid and acyl lipids profile of cold pressed corn oil

The major fatty acids of corn oil are specified as 59.8% linoleic acid, 25.8% oleic acid, 11.0% palmitic acid, 1.7% stearic acid, and 1.1% linolenic acid. A more detailed fatty acid composition range data is also available: myristic acid <0.1%, palmitic acid 8.0%–19.0%, palmitoleic acid <0.5%, stearic acid 0.5%–4.0%, oleic acid 19.0%–50.0%, linoleic acid 34.0%–62.0%, linolenic acid 0.1%–2.0%, arachidic acid <1.0%, gadoleic acid <0.5%, and behenic acids <0.5%. Deodorized corn oil contains 0.08%–0.12% of total tocopherols, which consist of 70%–80% α -, 20%–25% β -, and 3%–5% γ -tocopherols (Table 1). Likewise, around 1.3%–2.3% unsaponifiable matter is present and around 60% of this is sterols. The predominant sterols were β -sitosterol, campesterol, and stigmasterol. The ranges of tocopherols for corn oil are given as: α -tocopherol 116–172 ppm, β -tocopherol 0–22 ppm, γ -tocopherol 1119–1401 ppm, and δ -tocopherol 59–65 ppm. The most important physical properties of corn oil were given as: iodine value of 118–133, saponification number of 187–193, refractive index of 1.470–1.474, unsaponifiable content of 1.3%–2.3%, titer of 14–20 °C, melting point of –12 °C to –10 °C, and solidification point of –1.0 °C to –20.0 °C (O'Brien, 2004).

TABLE 1 Bioactive compounds and antioxidant activities of cold pressed corn (*Zea mays*) oil.

Compound	Amount	Reference
Antioxidant		
β -Carotene	3.56 mg/100 g	Rafalowski et al. (2008)
Chlorophylls content	4.9 mg/kg	Tuberoso et al. (2007)
TEAC activity	2.30 mmol/L	Tuberoso et al. (2007)
EC ₅₀ value	14.8 mg	Siger et al. (2008)
DPPH [•] radical scavenging activity	11.1%	Siger et al. (2008)
Total phenolic content	1.26 mg CAE/100 g	Siger et al. (2008)
Sterol and tocopherol		
β -Sitosterol	60%–62%	Stan (2015)
Campesterol	19%–22%	Stan (2015)
Stigmasterol	6%–7%	Stan (2015)
Sitosterol	2%–4%	Stan (2015)
α -Tocopherol	248.8 mg/kg	Rafalowski et al. (2008)
γ -Tocopherol	91.7 mg/100 g	Rafalowski et al. (2008)
δ -Tocopherol	5.93 mg/100 g	Rafalowski et al. (2008)
Phenolic compound		
Rutin	14.1 mg/kg	Pandey et al. (2013)
Hesperidin	23.8 mg/kg	Pandey et al. (2013)
Ferulic acid	0.5 ppm	Tuberoso et al. (2007)
<i>trans</i> -cinnamic acid	0.9 ppm	Tuberoso et al. (2007)
Vanillin	2.8 ppm	Tuberoso et al. (2007)
<i>p</i> -hydroxybenzoic acid	1.7 mg/100 g	Siger et al. (2008)
<i>p</i> -coumaric acid	1.9 mg/100 g	Siger et al. (2008)
Ferulic acid	5.8 mg/100 g	Siger et al. (2008)
Sinapic acid	0.6 mg/100 g	Siger et al. (2008)

4 Purification of cold pressed oil

In oil-bearing seeds, rendering cannot liberate oil from the cellular structure. The cell wall of seeds is broken by grinding, flaking, rolling, or pressing under high pressure to expel oil. The moisture content of the corn germs is adjusted to 12% by water conditioning before cold pressing, since that moisture level is determined as the best point for maximum oil yield and ease of the operation by preexperiments. Generally, the sequence of modern operation in cold pressing seeds is as follows: (1) to remove any stray bits of metal, seeds are passed over magnetic separators; (2) the hulls or shells are removed; (3) kernels are ground to convert them to coarse meal by hammer milling or by grinding them between grooved rollers; (4) finally, the material is pressed in screw or hydraulic presses without preliminary heating. Oil expelled without heating is often of edible quality without refining or further processing, and contains the least amount of impurities. Such oils are known as virgin, cold pressed, or cold drawn. Heating pressing of coarse meal removes more oil and also removes higher quantities of nonglyceride impurities like color bodies, phospholipids, and unsaponifiable matter. These oils are more highly colored than cold pressed oils (Aydeniz, Güneşer, & Yılmaz, 2014; Yılmaz, Arsunar, Aydeniz, & Güneşer, 2015).

5 Sterols and tocopherols composition of cold pressed corn oil

Table 1 presents the main bioactive compounds and antioxidant activities of cold pressed corn oil (CPCO). Some sterols (24-methylene cholesterol, campesterol, D-7-campesterol, D-5,23-stigmasterol, cholesterol, and D-5,24-stigmasterol) were present in trace amounts in the CPCO. A study of 15 samples found that the chief compounds present were β -sitosterol (60%–62%), campesterol (19%–22%), and stigmasterol (6%–7%). These values of given sterols agreed with the vegetable oils samples (Stan, 2015). The chief sterols of corn oils have been found to be β -sitosterol, campesterol, and stigmasterol (Strecker, Bieber, Maza, Grossberger, & Doskoczynski, 1996), whereas the quantification of tocopherol was as α -tocopherol in these samples. CPCO has a higher amount of α -tocopherol (248.8 mg/kg). A study was carried out to estimate the tocopherol from refined and CPCO with other oils (Rafalowski, Zegarska, Kuncewicz, & Borejszo, 2008). The results showed in cold pressed samples as 27.22, 91.76, and 5.93 mg/100 g oil of α -tocopherol, γ -tocopherol, and δ -tocopherol, respectively. In refined oils, these tocopherols were shown as 26.11, 44.31, and 0.0 mg/100 g oil, respectively. According to the findings of Strecker et al. (1996), the levels of tocopherols were 19.1, 94.2, and 4.2 mg/100 g for α -, γ -, and δ -tocopherols, respectively.

6 Phenolics composition

Studies did not report any phenolic compound in refined oil. This may be lost during refining or processing from corn oil. Eight different samples (baby corn, raw, boiled, baked, fried, ripe corns, popcorn, and stylar) were analyzed for phenolic content (Pandey, Singh, Maurya, Singh, & Singh, 2013). Only rutin (14.17 mg/kg) and hesperidin (23.81 mg/kg) were quantified among eight samples of oils. The amounts of phenolics in raw maize were found to be 4.6, 3.2, 0.99, and 0.57 mg/g of gallic, caffeic, ferulic, and salicylic acids, respectively. Tuberoso, Kowalczyk, Sarritzu, and Cabras (2007) reported that the maize oil contained 0.5 ppm ferulic acid, 0.9 ppm *trans*-cinnamic acid, and 2.8 ppm vanillin. There was quantification of phenolic acids in CPCO, and 1.7, 1.9, 5.8, and 0.6 mg/100 g oil of *p*-hydroxybenzoic, *p*-coumaric, ferulic, and sinapic acids, respectively, was found (Siger, Nogala-Kalucka, & Lampart-Szczapa, 2008). It was concluded from the previous literature that CPCO contained a number of phenolic contents. These phenolics are antioxidants, which may help to improve health status through antioxidant, antibacterial, antiinflammatory, antiallergic, antithrombotic, and anticarcinogenic activities, and hepatoprotective and vasodilatory actions (Pandey et al., 2013).

7 Volatiles and aromatics composition

There are 22 volatiles found in cold pressed corn oil and seven volatiles in hexane-extracted corn oil. Obviously, maximum volatiles are missing in the pure sample, probably through the alkali refining and deodorization processes. The rich volatiles found in the CPCO sample were hexanal, acetic acid, 3-methylbutanal, hexanoic acid, and butanoic acid. It has been identified that in any food product, the characteristic smell is provided by the volatiles found in combination, but not in ranges of maximum amounts. The definition of aroma threshold value is most minimal concentration of a substance that can be detected by a human nose. The aroma threshold of certain volatiles is considerably lower, which can account for the perception of aroma at lower amounts relative to other volatiles found in similar foods at maximum concentrations (Meilgaard, Civille, & Carr, 1991).

The descriptive characteristics of aromatic components in samples were green, fresh, herbal, and sour. Limited literature is available regarding volatile compounds of corn oil. There were 29 volatile components present in corn oil (Tuan, 2011). In another study, octane, 2,6-dimethyl-3-heptanone, 2-heptenal, 2-pentylfuran, nonanal, hexadecanoic acid, 9-octadecenoic acid ethyl ester, acetic acid, α -pinene, benzaldehyde, limonene, and 3-carene were determined.

Cold pressing corn germ contained many volatile compounds, wherein Solid Phase Microextraction (SPME) with different resins was used to quantify the volatiles in corn (Steenenson, Lee, & Min, 2002). The most abundant available volatile components identified at room temperature (storage 12–103 months) of corn oil were 2-heptenal, trans-2-octenal, pentane, and 2,4-heptedienal (Goicoechea & Guillén, 2014). Many secondary oxidized components such as alkanals, 2-alkenals, 2,4-alkadienals, acids, alcohols, ketones, alkylfurans, lactones, and a,b-unsaturated aldehydes were detected. With long duration, the type and concentration of volatile components were improved. At the initiation of the storage study, various resemblances between analyzed volatile components like hexanoic and butanoic acid, hexenal, nonanal, limonene, and some furans were determined.

8 Antioxidant activity of cold pressed corn oil

Table 1 presents the main bioactive compounds and antioxidant activities of cold pressed *Zea mays* oil. Various chemical compounds have been found in corn oils that showed strong importance in its nutritional potential. Various colored pigments and phenolics present in Zea oils have been lost during processing and during refining. The amount of the β -carotene content was found to be 91 ppm in commercial corn oil (Dauqan, Sani, Abdullah, Muhamad, & Top, 2011), while this content was present as 3.56 mg/100 g oil in CPCO, which is a negligible amount in processed or refined oil (Rafalowski et al., 2008). Furthermore, compounds like β -carotene (content 0.9 mg/kg), chlorophylls (content 4.9 mg/kg), and TEAC activity of 2.30 mmol/L were exploited for corn oil (Tuberoso et al., 2007). EC₅₀ values, DPPH-radical scavenging activity, and total phenolic content (as caffeic acid equivalent) were 14.8 mg, 11.1%, and 1.26 mg CAE/100 g oil, respectively (Siger et al., 2008).

9 Health benefits of cold pressed corn oil

Potential health benefits of corn oil include its capacity to lower inflammation, reduce oxidative stress, lower allergic reactions, reduce “bad” cholesterol, improve eye health, and protect the skin. Consumers usually use corn oil for cooking due to its flavor. However, some health-conscious consumers use organic corn oil in medicinal use and topical application due to presence of various health components. Cold pressed organic oil delivers many health benefits. Polyunsaturated and monounsaturated fatty acids are both required to reduce excess cholesterol and reduce inflammation. However, neither of these acids are balanced in refined oils (Rose, Thomson, & Williams, 1965). Monounsaturated fats and polyunsaturated fats are found in high quantities in unrefined corn oil, which includes oleic and linoleic acids, although when it comes to the balance of omega-6 and omega-3 to fatty acids in corn oil, the ratio is far from ideal. Omega-6 fatty acids were 50 times greater than omega-3 fatty acids in oil, whereas the recommended ratio is 1:1. Omega-6 and omega-3 are considered to be effective against inflammation and relieve symptoms of headaches, arthritis, and gastrointestinal problems.

Several compounds in corn oil are considered to lower allergic reactivity in the body, also helping to reduce the symptoms of rhinitis and asthma. Flavonoids, like lutein, can lower free radical activity in the body (Mulrooney & Grimble, 1993). Lutein is also recommended to improve eye health and reduce the development of cataracts. The tocopherols and antioxidants in corn oil could help to reduce skin infections and to prevent blemishes, irritation, psoriasis, and eczema. High levels of vitamin as well as limited levels of xanthins, vitamin A, and trace amounts of other antioxidant compounds can be beneficial for health (Staughton, 2019).

References

- Anonymous (2015). (2015). <http://www.worldofcorn.com/#world-cornproduction>. Accessed 26 August 2019.
- Aydeniz, B., Güneşer, O., & Yılmaz, E. (2014). Physico-chemical, sensory and aromatic properties of cold press produced safflower oil. *Journal of the American Oil Chemists' Society*, 91, 99–110. <https://doi.org/10.1007/s11746-013-2355-4>.
- Carrillo, W., Carpio, C., Morales, D., Vilcacundo, E., Álvarez, M., & Silva, M. (2017). Content of fatty acids in corn (*Zea mays* L.) oil from Ecuador. *Asian Journal of Pharmaceutical and Clinical Research*, 10(8), 150–153.
- Codex Stan. (2015). Codex Standard for named vegetable oils. pp. 1–13. http://www.fao.org/input/.../standards/336/CXS_210e_2015.pdf. Accessed 13 September 2019.
- Dauqan, E., Sani, H. A., Abdullah, A., Muhamad, H., & Top, A. G. (2011). Vitamin E and beta carotene composition in four different vegetable oils. *American Journal of Applied Sciences*, 8, 407–412. <https://doi.org/10.3844/ajassp.2011.407.412>.

- Goicoechea, E., & Guillén, M. D. (2014). Volatile compounds generated in corn oil stored at room temperature. Presence of toxic compounds. *European Journal of Lipid Science and Technology*, 116, 395–406.
- Güneşer, B. A., Yılmaz, E., & Ok, S. (2017). Cold pressed versus refined winterized corn oils: Quality, composition and aroma. *Grasas y Aceites*, 68(2), e194.
- Khoddami, A., Che Man, Y. B., & Roberts, T. H. (2014). Physicochemical properties and fatty acid profile of seed oils from pomegranate (*Punica granatum* L.) extracted by cold pressing. *European Journal of Lipid Science and Technology*, 116, 553–562. <https://doi.org/10.1002/ejlt.201300416>.
- Meilgaard, M., Civille, G. V., & Carr, B. T. (1991). *Sensory evaluation techniques*. Boca Raton, FL: CRC Press ISBN 10: 0849342805/ISBN 13: 9780849342806.
- Mulrooney, H. M., & Grimble, R. F. (1993). Influence of butter and of corn, coconut and fish oils on the effects of recombinant human tumor necrosis factor- α in rats. *Clinical Science*, 84(1), 105–112.
- O'Brien, R. D. (2004). *Fats and oils: Formulating and processing for applications*. New York: CRC Press. ISBN 9780203483664.
- Pandey, R., Singh, A., Maurya, S., Singh, U. P., & Singh, M. (2013). Phenolic acids in different preparations of maize (*Zea mays*) and their role in human health. *International Journal of Current Microbiology and Applied Sciences*, 2, 84–92 (ISSN: 2319-7706).
- Rafalowski, R., Zegarska, Z., Kuncewicz, A., & Borejszo, Z. (2008). Fatty acid composition, tocopherol and β -carotene content in Polish commercial vegetable oils. *Pakistan Journal of Nutrition*, 7, 278–282.
- Ray, K., Banerjee, H., Dutta, S., Hazra, A. K., & Majumdar, K. (2019). Macronutrients influence yield and oil quality of hybrid maize (*Zea mays* L.). *PLoS One*, 14(5), e0216939.
- Rose, G. A., Thomson, W. A., & Williams, R. T. (1965). Corn oil in treatment of Ischaemic heart disease. *British Medical Journal*, 1(5449), 1531–1533.
- Siger, A., Nogala-Kalucka, M., & Lampart-Szczapa, E. (2008). The content and antioxidant activity of phenolic compounds in cold pressed plant oils. *Journal of Food Lipids*, 15, 137–149.
- Staughton, J. (2019). Benefits & side effects of corn oil. <https://www.organicfacts.net/health-benefits/oils/corn-oil.html>.
- Steenson, D. F., Lee, J. H., & Min, D. B. (2002). Solid phase microextraction of volatile soybean oil and corn oil compounds. *Journal of Food Science*, 67, 71–76. <https://doi.org/10.1111/j.1365-2621.2002.tb11361.x>.
- Strecker, L. R., Bieber, M. A., Maza, A., Grossberger, T., & Doskoczynski, W. J. (1996). Corn oil. In Y. H. Hui (Ed.), *Bailey's industrial oil and fat products* (pp. 125–158). New York: Wiley ISBN-13: 978-0471594307/ISBN-10: 047159430X.
- Tuan, H. Q. (2011). *Characterization of natural edible oils regarding their quality and safety related constituents*. Master Thesis Division of Food Chemistry, Department of Food Sciences and Technology, University of Natural Resources and Life Sciences, Vienna, Austria.
- Tuberoso, C. I. G., Kowalczyk, A., Sarritzu, E., & Cabras, P. (2007). Determination of antioxidant compounds and antioxidant activity in commercial oilseeds for food use. *Food Chemistry*, 103, 1494–1501.
- Yılmaz, E., Arsunar, E. S., Aydeniz, B., & Güneşer, O. (2015). Cold pressed capia pepperseed (*Capsicum annuum* L.) oils: Composition, aroma and sensory properties. *European Journal of Lipid Science and Technology*, 117, 1016–1026.
- Yılmaz, E., Aydeniz, B., Güneşer, O., & Arsunar, E. S. (2015). Sensory and physico-chemical properties of cold press produced tomato (*Lycopersicon esculentum* L.) seed oils. *Journal of the American Oil Chemists' Society*, 92, 833–842. <https://doi.org/10.1007/s11746-015-2648-x>.